Recursion Master Practice Sheet

1. Too Easy (Absolute Basics)

- Print Numbers from 1 to N

  Write a recursive function to print numbers from 1 to N.

- Print Numbers from N to 1

  Write a recursive function to print numbers from N to 1.

- Sum of First N Natural Numbers

  Find the sum of the first N numbers using recursion.

- Factorial of a Number

  Calculate factorial using recursion.

- Check Even or Odd

  Check if a number is even or odd using recursion.

- Multiply Two Numbers using Recursion

  Use recursion to multiply two integers.

2. Easy (Arrays and Numbers)

- Count Digits of a Number

  Count the number of digits in a given integer using recursion.

Recursion Master Practice Sheet

- Sum of Digits

  Calculate the sum of digits of a number using recursion.

- Print Array Elements

  Print all elements of an array recursively.

- Find Maximum in Array

  Find the maximum element in an array using recursion.

- Find Minimum in Array

  Find the minimum element in an array using recursion.

- Linear Search in Array

  Implement linear search in an array using recursion.

- Binary Search in Array

  Implement binary search using recursion.

- Check if Array is Sorted

  Check if an array is sorted using recursion.

- Reverse Print Array

  Print elements of an array in reverse order using recursion.

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- Sum of Array Elements

  Find sum of elements in an array using recursion.

3. Medium (Logic Building and Strings)

- Reverse a Number

  Reverse a number using recursion.

- Palindrome Check (Number/String)

  Check if a number or string is a palindrome.

- Binary Representation of a Number

  Print binary representation of a number using recursion.

- Print String in Reverse

  Print a string in reverse using recursion.

- Count Vowels in String

  Count the number of vowels in a string using recursion.

- Replace Character in String

  Replace all occurrences of a character in a string.

- Remove Duplicates from String

  Remove duplicate characters from a string using recursion.

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- Check Balanced Parentheses

  Check if parentheses are balanced using recursion.

4. Hard (Classic Recursion Problems)

- Fibonacci Number

  Calculate the N-th Fibonacci number using recursion.

- Power Function

  Implement power(a, b) using recursion.

- Print All Subsets of a String

  Generate all subsets of a string using recursion.

- Generate All Permutations of a String

  Print all permutations of a given string.

- String to Integer Conversion

  Convert a numeric string to an integer using recursion.

5. Very Hard (Advanced Challenges)

- Subset Sum Problem

  Find all subsets that sum to a specific value.

Recursion Master Practice Sheet

- Tower of Hanoi

  Solve Tower of Hanoi problem for N disks.

- N-th Catalan Number

  Calculate N-th Catalan number using recursion.

- Count Paths in Maze (Backtracking)

  Count all possible paths in a maze from (0,0) to (n-1,m-1).

- Word Break Problem

  Given a dictionary and a string, check if it can be segmented.